

# Automata

# languages lexers, parsers, interpreters



# Rekindle chapter 1 and 2

An Introduction to Formal Languages and  
Automata 6th Edition

# Grammar

- $G = (V, T, S, P)$ ,
  - $V$  is a finite set of objects called variables,
  - $T$  is a finite set of objects called terminal symbols,
  - $S \in V$  is a special symbol called the start variable,
  - $P$  is a finite set of productions.

The set of all such terminal strings is the language defined or generated by the grammar

# Excercise

Let  $L = \{ab, aa, baa\}$ .

Which of the following strings are in  $L^*$ :

1. abaabaaabaa
2. aaaabaaaa
3. baaaaabaaaab
4. baaaaabaa?

Which strings are in  $L^4$ ?

# Exercise

Give a simple description of the language generated by the grammar with productions generate?

- $S \rightarrow aaA,$
- $A \rightarrow bS,$
- $S \rightarrow \lambda.$

# Exercise

Give a simple description of the language generated by the grammar with productions generate?

- $S \rightarrow Aa$ ,
- $A \rightarrow B$ ,
- $B \rightarrow Aa$ .

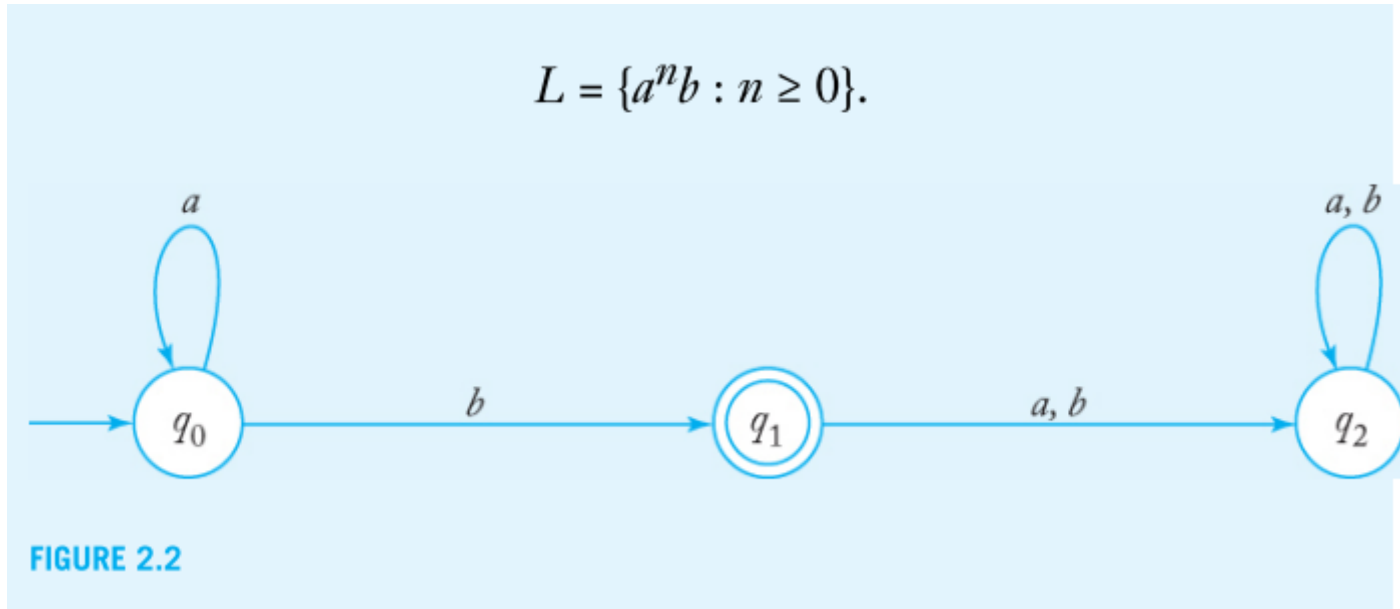
# Definition

A deterministic finite **accepter** or **dfa** is defined by the quintuple  $M = (Q, \Sigma, \delta, q_0, F)$ ,

where

$Q$  is a finite set of **internal states**,  $\Sigma$  is a finite set of symbols called the **input alphabet**,  $\delta : Q \times \Sigma \rightarrow Q$  is a total function called the **transition function**,  $q_0 \in Q$  is the **initial state**,  $F \subseteq Q$  is a set of **final states**.

# Transition graph





# Definition

A language  $L$  is called **regular** if and only if there exists some deterministic finite acceptor  $M$  such that

$$L = L(M).$$

# Exercise

For  $\Sigma = \{a, b\}$ , construct dfa's that accept the sets consisting of

- (a) all strings of even length.
- (b) all strings of length greater than 5.
- (c) all strings with an even number of a's.